# Storage Lab: LVM Physical Volume Resize & Logical Volume Online Extension

**Duration : 30 Min.**

## Scenario

An underlying disk device `/dev/sdb` has been enlarged from 50 GB to 100 GB in the hypervisor/cloud console. However, the logical volume `/dev/vg_data/lv_app` mounted at `/data` still displays 50 GB. The administrator must resize the physical volume, extend the logical volume to consume all newly available volume group space, grow the EXT4 filesystem online without unmounting `/data`, and record the expansion metrics in `lvm_resize_report.json`.

## Task Objectives

Perform the following LVM volume extension tasks:

> **Pre-Staged Files in `student_workspace/`:**
> • **`lvm_resize_report.json`**: Starter LVM online extension report with `TODO_*` placeholders.

1. Inspect the underlying block device `/dev/sdb` to detect the newly expanded physical disk capacity.
2. Update the LVM metadata via `pvresize` so the physical volume and volume group recognize the unallocated space (100 GB).
3. Extend the logical volume `/dev/vg_data/lv_app` to consume 100% of the newly available volume group extents (`lvextend`).
4. Grow the live filesystem online (`resize2fs`) without unmounting the volume or interrupting service operations.
5. Record your executed commands, detected capacities, and resize completion status in `lvm_resize_report.json`.

## Instructions to Perform the Task

1. Open your assessment **Terminal** or storage management CLI interface.
2. Execute the required storage management commands, volume operations, or configuration edits.
3. Update `lvm_resize_report.json` inside `student_workspace/` replacing all `TODO_*` placeholders with your verified configuration values.

4. Verify your storage configurations locally by executing:

```
python run.py
```

5. Once verified, return to the platform dashboard and click **"Run Tests"** / **"Full Eval"** to evaluate your score!

## Validation

Once your storage tasks are executed, return to the platform dashboard and click the **"Run Tests"** button. This will automatically evaluate your changes and generate your score!

## Grading Criteria

Your performance will be evaluated based on the following test cases:

| Test Case | Requirement | Marks |
|---|---|---|
| **TC1** | PV resize: pvresize successfully detects expanded underlying block device capacity | 2 Marks |
| **TC2** | VG metadata: volume group free extents updated and verified via vgs | 2 Marks |
| **TC3** | LV extension: logical volume extended to take 100% of newly available extents | 3 Marks |
| **TC4** | Filesystem growth: resize2fs completes online with zero data loss and df confirms 100GB usable space | 3 Marks |

**Total Score: 10 Marks**

## Important Notes

- This is an **Auto-Evaluated** storage assessment.
- Ensure all created volumes, files, and scripts are properly configured and saved before submitting.